Point of Sale System — Combined Conceptual Schema

Mini-World Description

Our mini-world is a point of sale (POS) system for a games and collectibles store. The store sells products that include video games, manga, books, trading cards, figures, collectibles, apparel, and gaming accessories. Our system keeps track of products, categories, stock levels, suppliers, discounts, customers, employees, reviews, and sale transactions.

Employees must log in before using the system; each employee is either a Manager or a Cashier. Managers supervise other employees and are the only ones who can update inventory, edit inventory, change prices, delete products, and edit discounts. Employees use a Virtual Register to check out customers, where they can input any item, add or delete items in the current transaction, and see the count of each individual item.

Customers can purchase multiple products in one transaction. The system keeps record of customer information like name, email, and loyalty points.

Products are items sold by the store, such as video games, manga, collectibles, etc. The system keeps track of the price, SKU, condition, and stock information.

Categories are used to organize products into groups such as consoles, discs, merch, and accessories. Each product belongs to exactly one category and has information such as the name, description, price, condition, and reorder threshold.

Suppliers are the ones who provide the products to the store. The system stores their contact information so the store can keep track of where and when the product came from.

A customer can make many transactions and have many items on that transaction. The system records the date, time, subtotal, tax, total, and payment information for each transaction. Transaction items represent the individual products included in a transaction, recording the product, quantity purchased, unit price, and line total.

The store also runs sale events, which group discounts under a name and a date range, and a rewards program, which defines how customers earn and redeem loyalty points and which tier they belong to. The system produces reports on sales, inventory, employee performance, sale events, and the rewards program, drawing on the transaction, inventory, and loyalty data it already collects.

Entities and Attributes

23 entities in total: 9 from the original team draft (with some attributes added or corrected) and 14 new ones.

EMPLOYEE

Employee_id (PK), First_name, Last_name, Email, Phone_number, Role, Hire_date, Password_hash, Supervisor_id (FK => EMPLOYEE, optional)

CUSTOMER

Customer_id (PK), First_name, Last_name, Email, Loyalty_points, Street#, Apt#, City, State, Zip_code, Created_date, Referred_by_id (FK => CUSTOMER, optional), Tier_id (FK => REWARD_TIER, optional)

CUSTOMER_PHONE

Customer_id (FK => CUSTOMER, part of PK), Phone_number (part of PK)

PRODUCT

Product_id (PK), SKU, Name, Description, Cost_price, Sell_price, Stock_quantity, Reorder_threshold, Condition, Category_id (FK => CATEGORY), Discount_id (FK => DISCOUNT, optional)

CATEGORY

Category_id (PK), Name, Description, Discount_id (FK => DISCOUNT, optional)

SUPPLIER

Supplier_id (PK), Name, Email, Phone, Street#, Apt#, City, State, Zip_code

SUPPLIER_PRODUCT

Supplier_id (FK => SUPPLIER, part of PK), Product_id (FK => PRODUCT, part of PK), Supplier_sku, Unit_cost, Lead_time_days

SUPPLIER_ORDER

Order_id (PK), Supplier_id (FK => SUPPLIER), Employee_id (FK => EMPLOYEE), Order_date, Received_date, Status

SUPPLIER_ORDER_ITEM

Order_id (FK => SUPPLIER_ORDER, part of PK), Product_id (FK => PRODUCT, part of PK), Quantity, Unit_cost

TRANSACTION

Transaction_id (PK), Customer_id (FK => CUSTOMER, optional), Employee_id (FK => EMPLOYEE), Register_id (FK => REGISTER), Date, Subtotal, Tax_amount, Total, Status (open, completed, voided)

TRANSACTION_ITEM

Transaction_id (FK => TRANSACTION, part of PK), Product_id (FK => PRODUCT, part of PK), Quantity, Unit_price, Line_total, Discount_id (FK => DISCOUNT, optional), Discount_amount

PAYMENT

Payment_id (PK), Transaction_id (FK => TRANSACTION), Method (credit, debit, cash), Amount, Card_last4, Change_given, Paid_at

DISCOUNT

Discount_id (PK), Name, Type, Code, Amount, Amount_type (percent, fixed), Start_date, End_date, Sale_event_id (FK => SALE_EVENT, optional)

SALE_EVENT

Sale_event_id (PK), Name, Description, Start_date, End_date, Is_active

REVIEW

Review_id (PK), Rating, Comment, Review_date, Product_id (FK => PRODUCT), Customer_id (FK => CUSTOMER), Transaction_id (FK => TRANSACTION, optional)

REGISTER

Register_id (PK), Register_number, Is_active

LOW_STOCK_ALERT

Alert_id (PK), Product_id (FK => PRODUCT), Triggered_at, Resolved_at, Order_id (FK => SUPPLIER_ORDER, optional)

INVENTORY_ADJUSTMENT

Adjustment_id (PK), Product_id (FK => PRODUCT), Employee_id (FK => EMPLOYEE), Quantity_change, Reason, Adjusted_at

PRICE_CHANGE

Price_change_id (PK), Product_id (FK => PRODUCT), Employee_id (FK => EMPLOYEE), Old_price, New_price, Changed_at

TAX_RATE

Tax_rate_id (PK), Category_id (FK => CATEGORY), Rate, Effective_from, Effective_to

LOYALTY_TRANSACTION

Loyalty_txn_id (PK), Customer_id (FK => CUSTOMER), Transaction_id (FK => TRANSACTION, optional), Program_id (FK => REWARDS_PROGRAM), Points_change, Reason, Created_at

REWARDS_PROGRAM

Program_id (PK), Name, Points_per_dollar, Redemption_rate, Effective_from, Effective_to

REWARD_TIER

Tier_id (PK), Program_id (FK => REWARDS_PROGRAM), Name, Min_points, Perk_description

Relationships

EMPLOYEEs process TRANSACTIONs — 1:N

CUSTOMERs make TRANSACTIONs — 1:N

TRANSACTIONs contain TRANSACTION_ITEMs — 1:N

PRODUCTs appear in TRANSACTION_ITEMs — 1:N

CATEGORYs classify PRODUCTs — 1:N

SUPPLIERs supply PRODUCTs — M:N (via SUPPLIER_PRODUCT)

EMPLOYEEs place SUPPLIER_ORDERs — 1:N

SUPPLIERs receive SUPPLIER_ORDERs — 1:N

EMPLOYEEs (managers) supervise EMPLOYEEs — 1:N

DISCOUNTs apply to CATEGORYs — 1:N

DISCOUNTs apply to PRODUCTs — 1:N

DISCOUNTs apply to TRANSACTION_ITEMs — 1:N

CUSTOMERs refer CUSTOMERs — 1:N

CUSTOMERs make REVIEWs — 1:N

PRODUCTs receive REVIEWs — 1:N

TRANSACTIONs may be the subject of REVIEWs — 1:N

CUSTOMERs have CUSTOMER_PHONEs — 1:N

SUPPLIER_ORDERs contain SUPPLIER_ORDER_ITEMs — 1:N

PRODUCTs appear in SUPPLIER_ORDER_ITEMs — 1:N

TRANSACTIONs are paid through PAYMENTs — 1:N

REGISTERs process TRANSACTIONs — 1:N

PRODUCTs trigger LOW_STOCK_ALERTs — 1:N

SUPPLIER_ORDERs resolve LOW_STOCK_ALERTs — 1:N

EMPLOYEEs make INVENTORY_ADJUSTMENTs — 1:N

PRODUCTs have INVENTORY_ADJUSTMENTs — 1:N

EMPLOYEEs make PRICE_CHANGEs — 1:N

PRODUCTs have PRICE_CHANGEs — 1:N

CATEGORYs have TAX_RATEs — 1:N

CUSTOMERs earn LOYALTY_TRANSACTIONs — 1:N

TRANSACTIONs generate LOYALTY_TRANSACTIONs — 1:N

REWARDS_PROGRAMs govern LOYALTY_TRANSACTIONs — 1:N

REWARDS_PROGRAMs define REWARD_TIERs — 1:N

REWARD_TIERs classify CUSTOMERs — 1:N

SALE_EVENTs group DISCOUNTs — 1:N

Constraints

Every entity's id is unique (primary key).

Product prices cannot be negative.

Product stock cannot be negative.

Reorder threshold cannot be negative.

Every product must belong to exactly one valid category.

Every transaction item must reference an existing product and belong to an existing transaction.

Transaction item quantity must be greater than 0.

Every transaction must be processed by exactly one employee and one register.

A transaction may be made by at most one customer.

A product cannot be sold if its stock is lower than the quantity requested.

Selling a product decreases its stock by the quantity sold, and only once the transaction's status is completed.

Items can be added to or removed from a transaction only while its status is open.

A voided transaction leaves stock unchanged.

Transaction total cannot be negative.

Every review must be made by an existing customer, about an existing product, and optionally an existing transaction.

Review rating must be between 1 and 5 stars.

The employee role must be either Manager or Cashier.

An employee's supervisor must hold the Manager role and cannot be the employee themselves.

A customer cannot refer themselves.

Only managers can edit inventory, prices, and discounts; this is enforced in the application layer.

Marking a supplier order received increases each ordered product's stock by the ordered quantity.

Supplier order item quantity must be greater than 0, and unit cost cannot be negative.

A sale event's end date cannot be earlier than its start date.

Discount end date cannot be earlier than its start date, and discount amount cannot be negative.

A reward tier's minimum points must be unique within its program, and higher tiers require more points than lower ones.

A customer's tier must match a tier whose minimum points is at or below the customer's current point total; this is enforced in the application layer.

A tax rate's effective end date, if any, cannot be earlier than its effective start date.